

Introduction to Amazon Identity and Access Management

Hands-on Lab

Table of Contents

INTRODUCTION TO AMAZON IDENTITY AND ACCESS MANAGEMENT	1
HANDS-ON LAB.....	1
LAB OVERVIEW	3
TOPICS COVERED	3
TASK 1: EXPLORE THE USERS AND GROUPS BUSINESS SCENARIO	3
TASK 2: ADD USERS TO GROUPS	8
BUSINESS SCENARIO	8
TASK 3: SIGN-IN AND TEST USERS	11

Lab Overview

In this lab, I explored how to add users to existing user groups in AWS. AWS Identity and Access Management (IAM) is a web service that allows AWS customers to manage users and user permissions in that environment. With IAM, it is possible to manage users, security credentials, access keys, and permissions (AWS, n.d).

Topics covered

- Exploring pre-created IAM users and Groups in the console
- Inspecting IAM policies attached to pre-created groups
- Adding users to groups with specific capabilities, as in a real-world scenario
- Locating and using the IAM sign-in URL
- Experimenting with the effects of policies on service access

Task 1: Explore the Users and Groups Business Scenario

In this task, I explored the users and groups that were already created in the IAM.

- Step 1: Search for IAM in the search bar
- Step 2: Choose users (user-1, user-2, user-3)
- Step 3: Open user-1 page, and notice the permissions tab
- Step 4: Open the group tab
- Step 5: Choose the “Security credentials” tab
- Step 6: Notice the existing groups (EC2-Admin, EC2-Support, S3-Support)

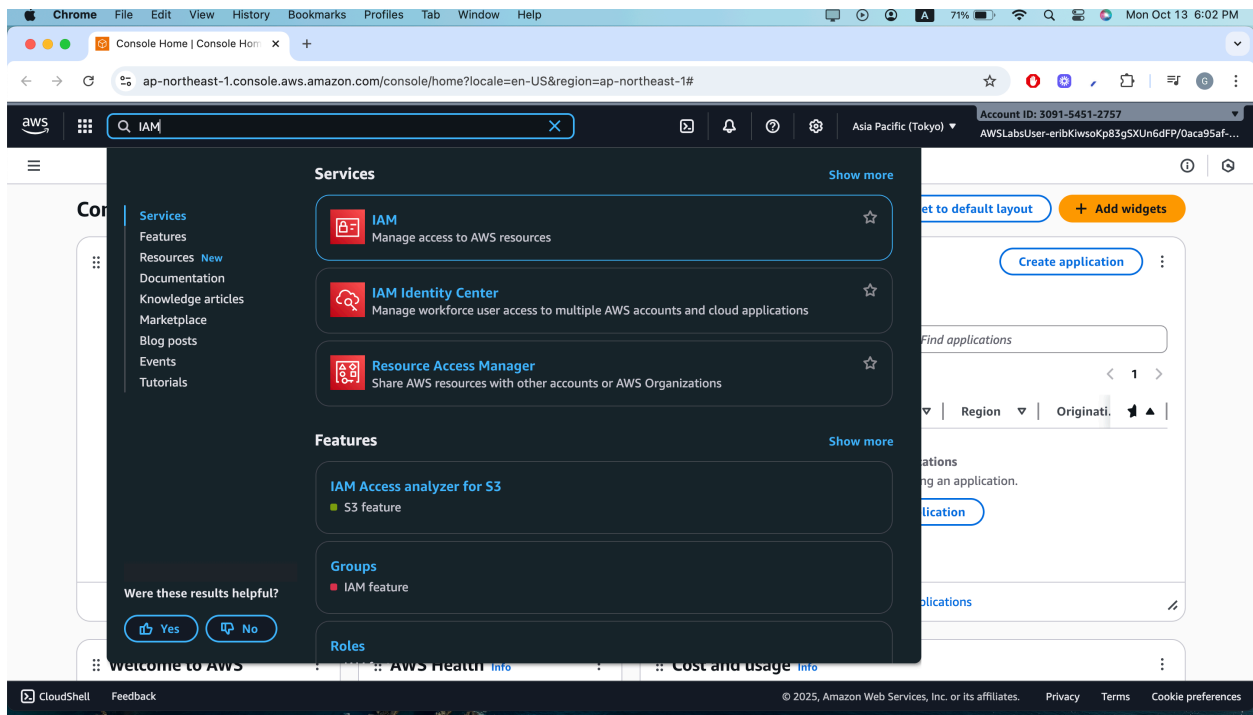


Figure 1: IAM search (source: personal collection)

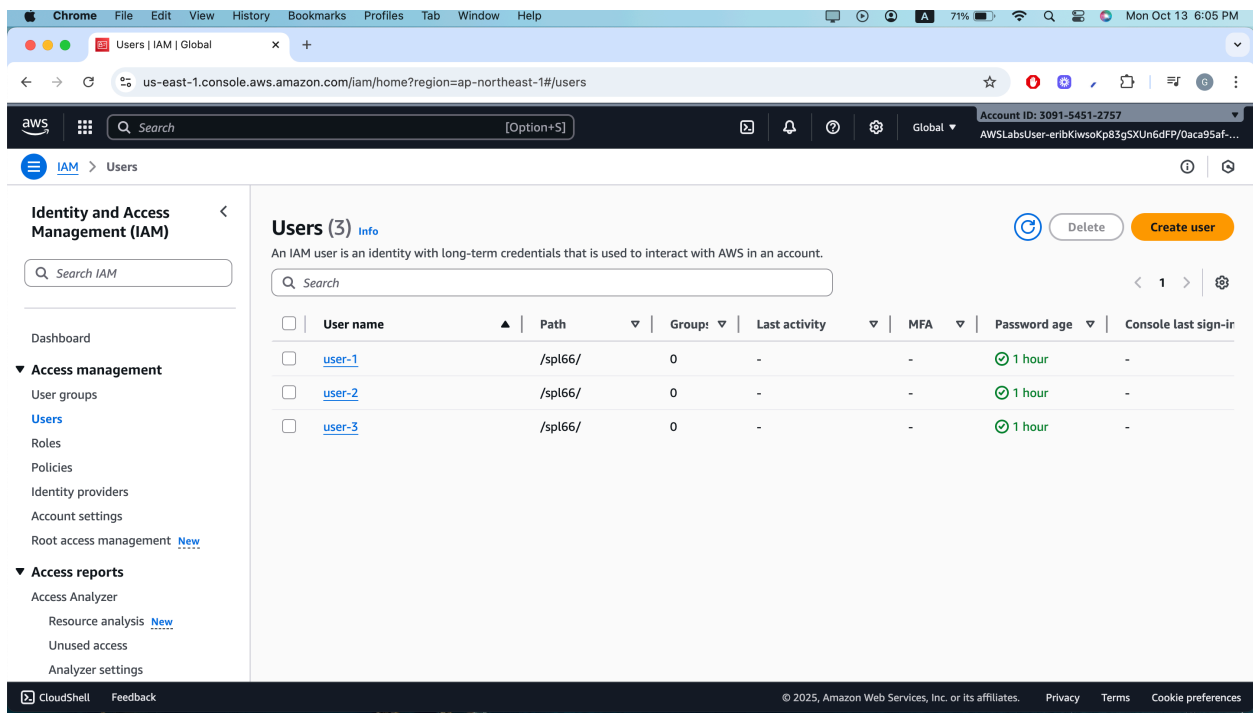


Figure 2: Users tab (source: personal collection)

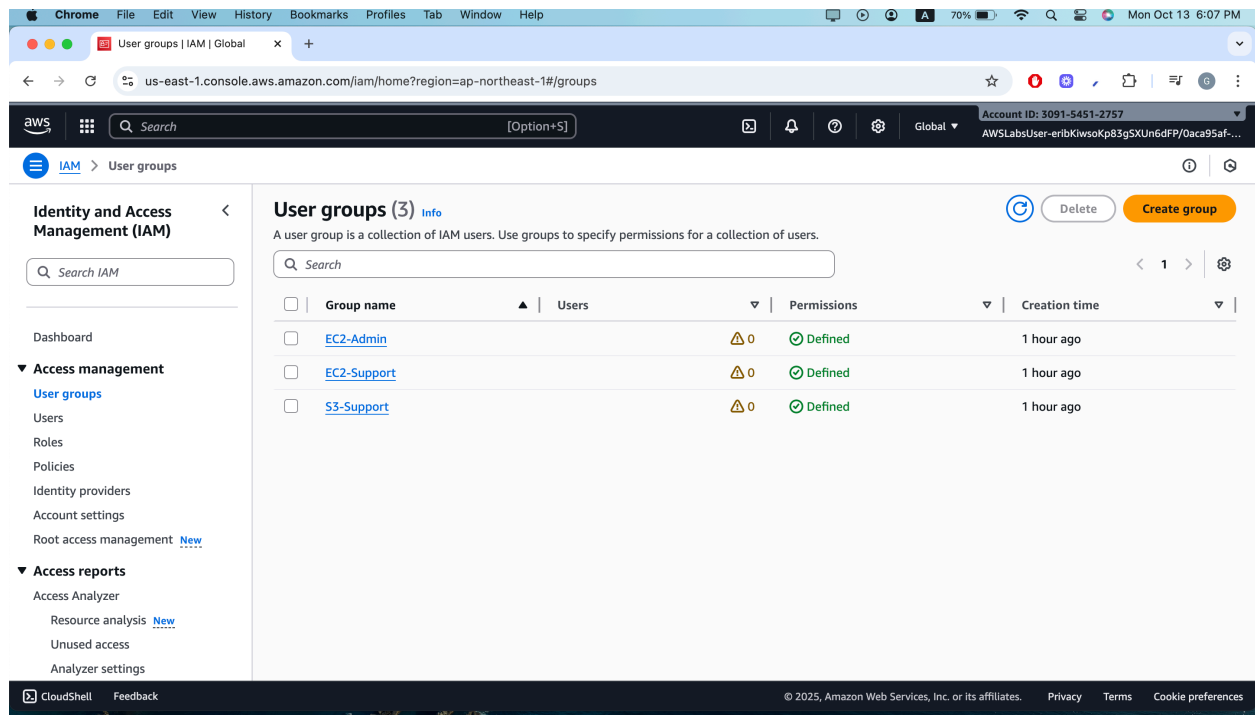


Figure 3: User groups (source: personal collection)

- Step 7: Choose the “Permissions” tab
- Step 8: Notice “AmazonEC2ReadOnlyAccess” policy
- Step 9: Notice “AmazonS3ReadOnlyAccess” policy
- Step 10: Notice “AmazonEC2ReadOnlyAccess” policy

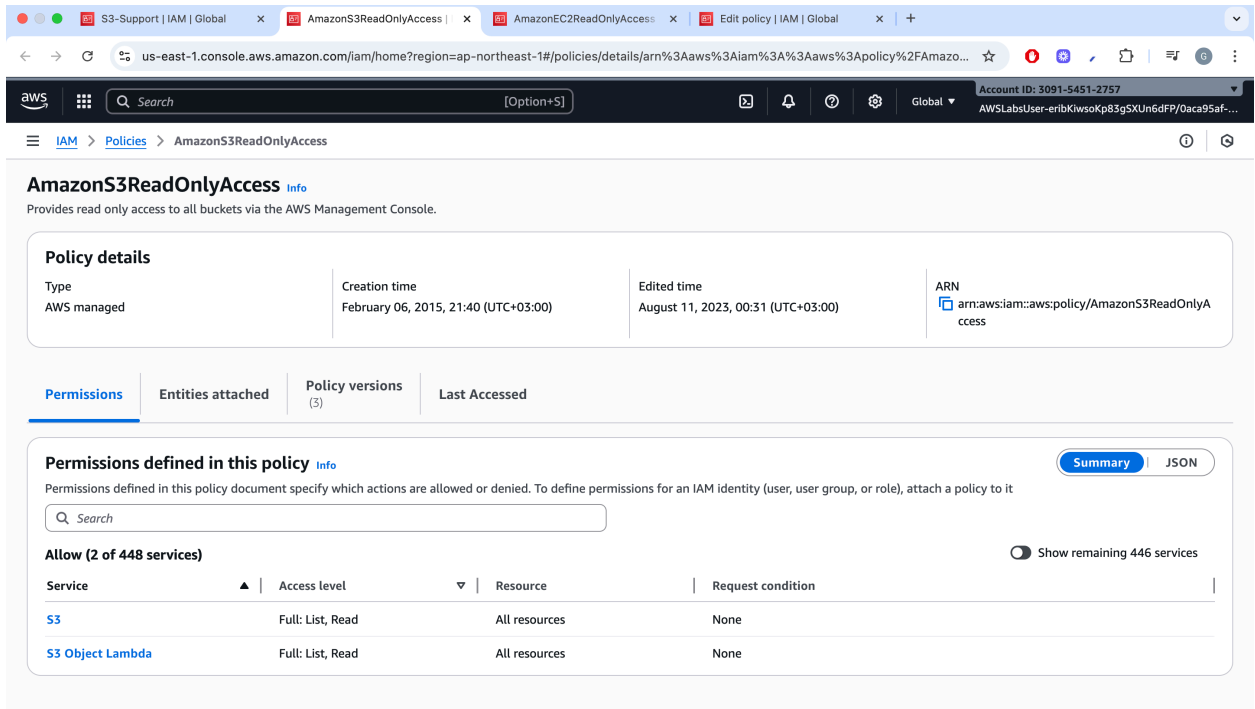


Figure 4: AmazonS3ReadOnlyAccess (source: personal collection)

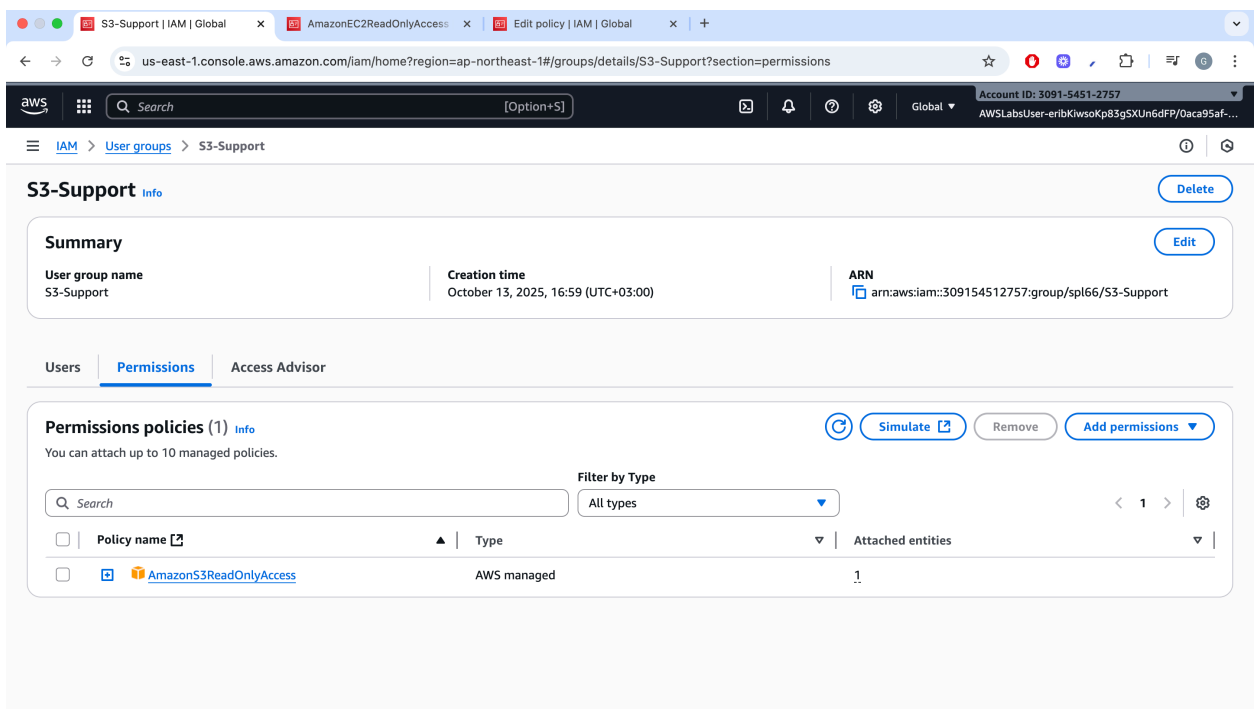


Figure 5: S3-Support permission tab (source: personal collection)

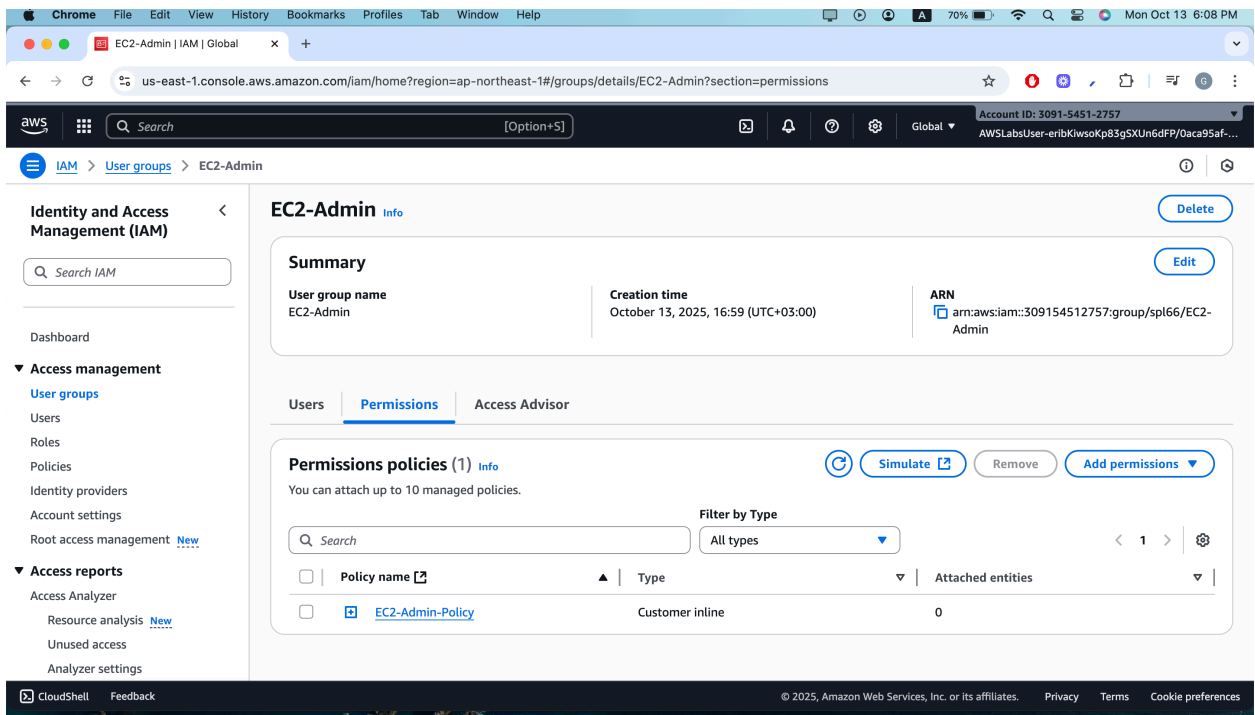


Figure 6: EC2-Admin permission tab (source: personal collection)

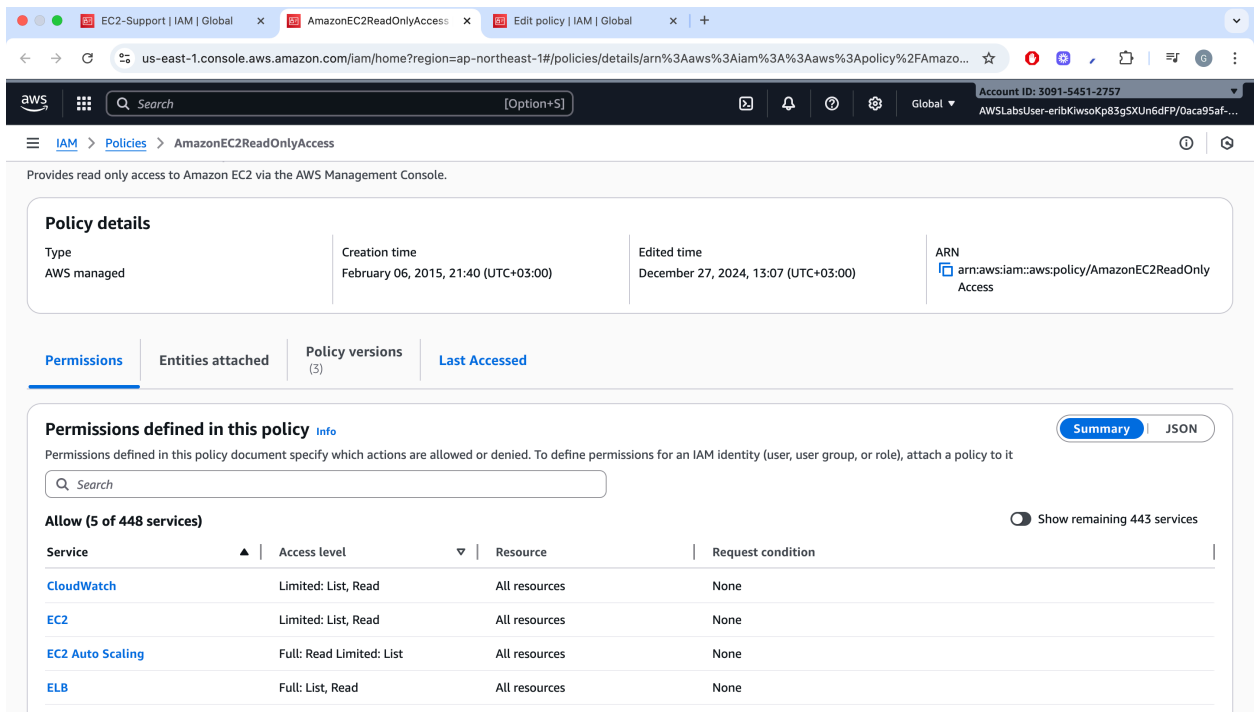


Figure 7: Policy details (source: personal collection)

Task 2: Add Users to Groups

Business Scenario

Your company is growing its use of Amazon Web Services, using many Amazon EC2 instances and a great deal of Amazon S3 storage. These are added as follows:

User	In Group	Permissions
User-1	S3-support	Read-Only access to Amazon S3
User-2	EC2-Support	Read-Only access to Amazon EC2
User-3	EC2-Admin	View, Start and Stop Amazon EC2 instances

- Step 1: Add user-1 to the S3-Support Group
- Step 2: Add user-2 to the EC2-Support Group
- Step 3: Add user-3 to the EC2-admin Group

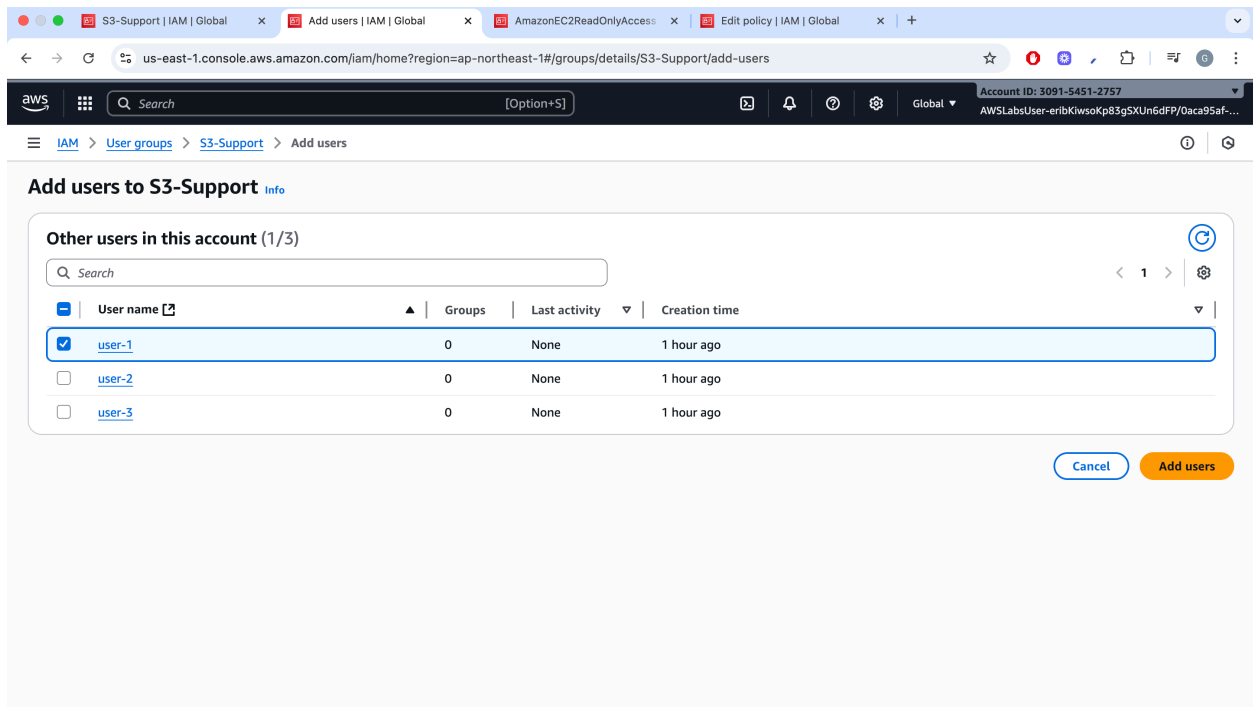


Figure 8: Adding user-1 to S3-Support (source: personal collection)

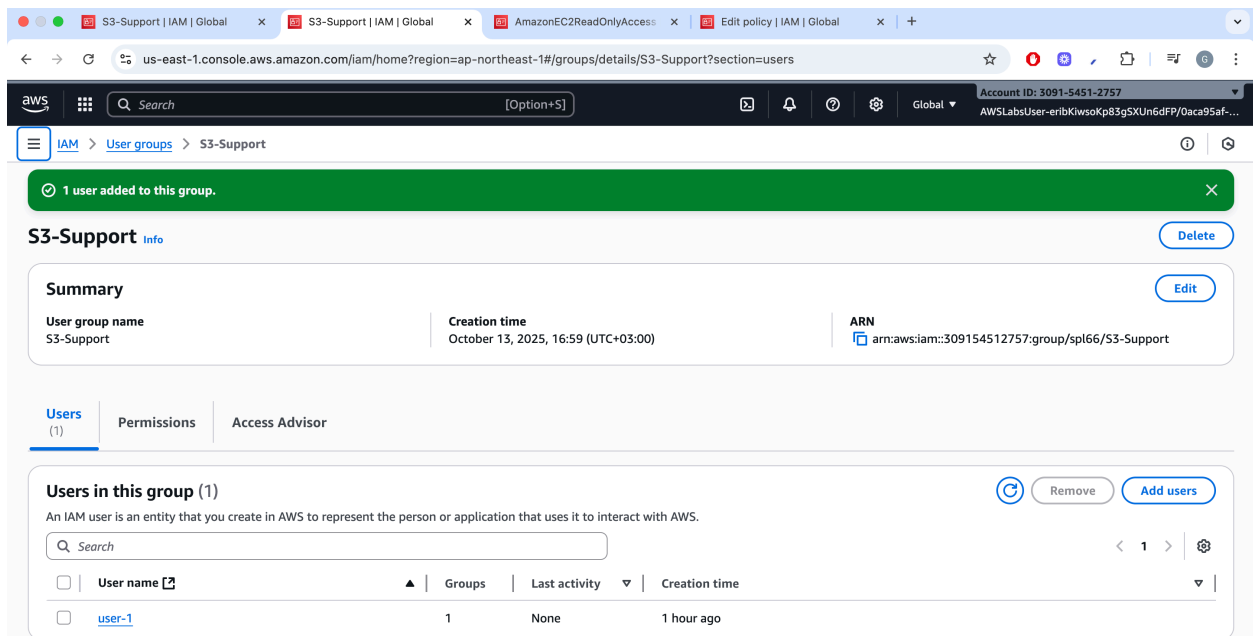


Figure 9: S3-Support group overview (source: personal collection)

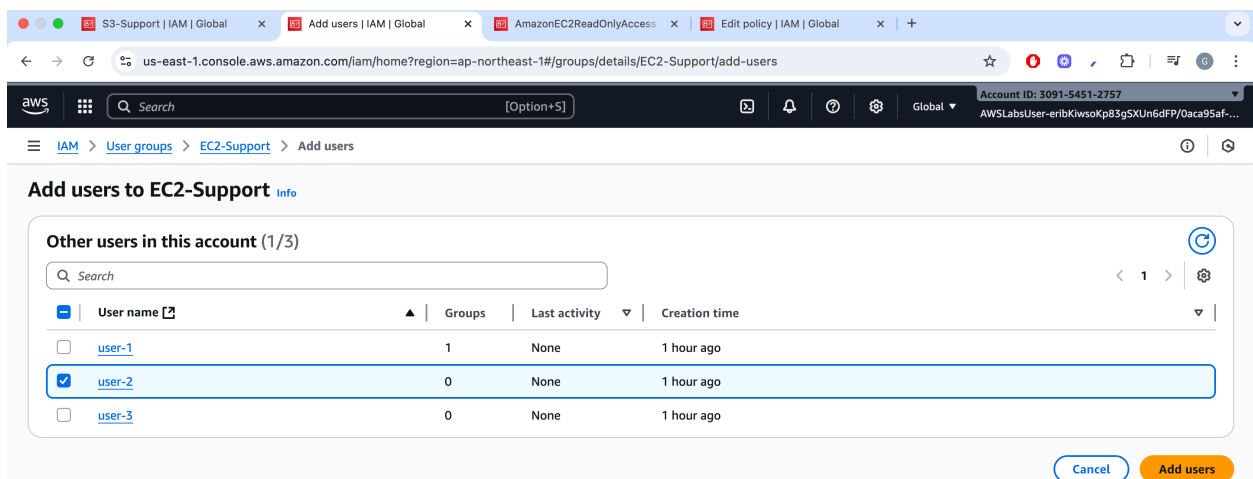


Figure 10: Adding user-2 to EC2-Support (source: personal collection)

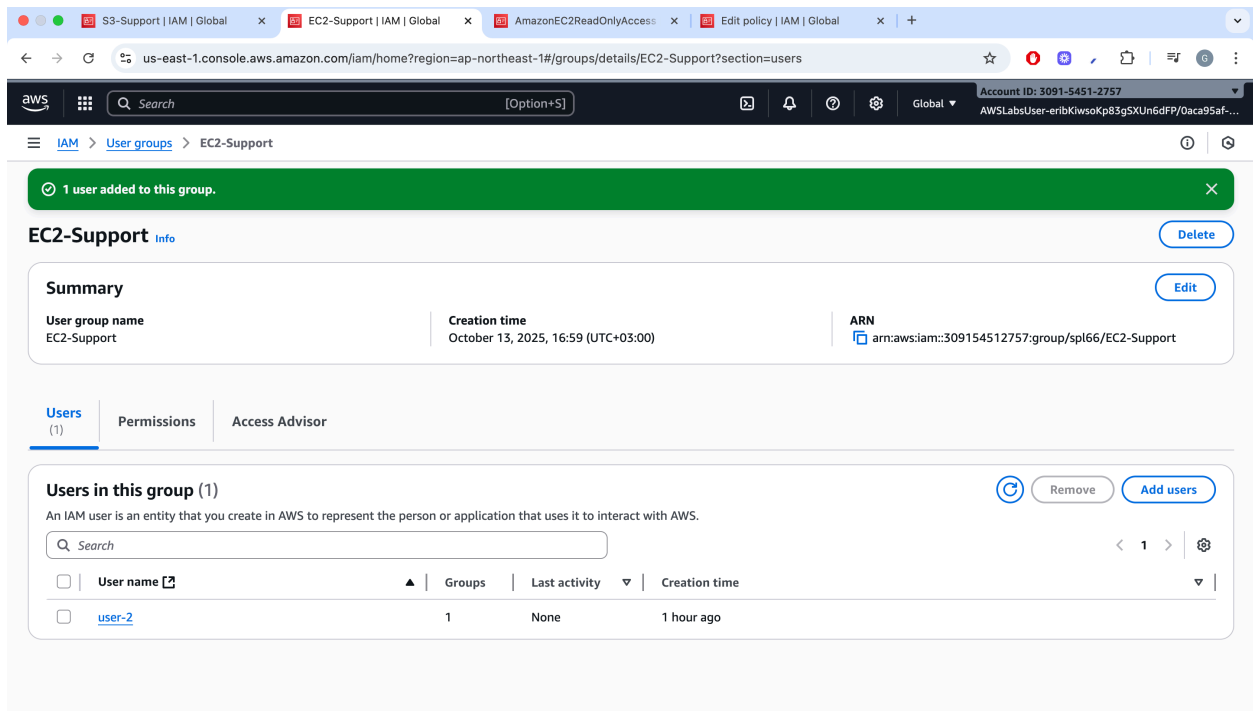


Figure 11: Adding user-3 to EC2-Support (source: personal collection)

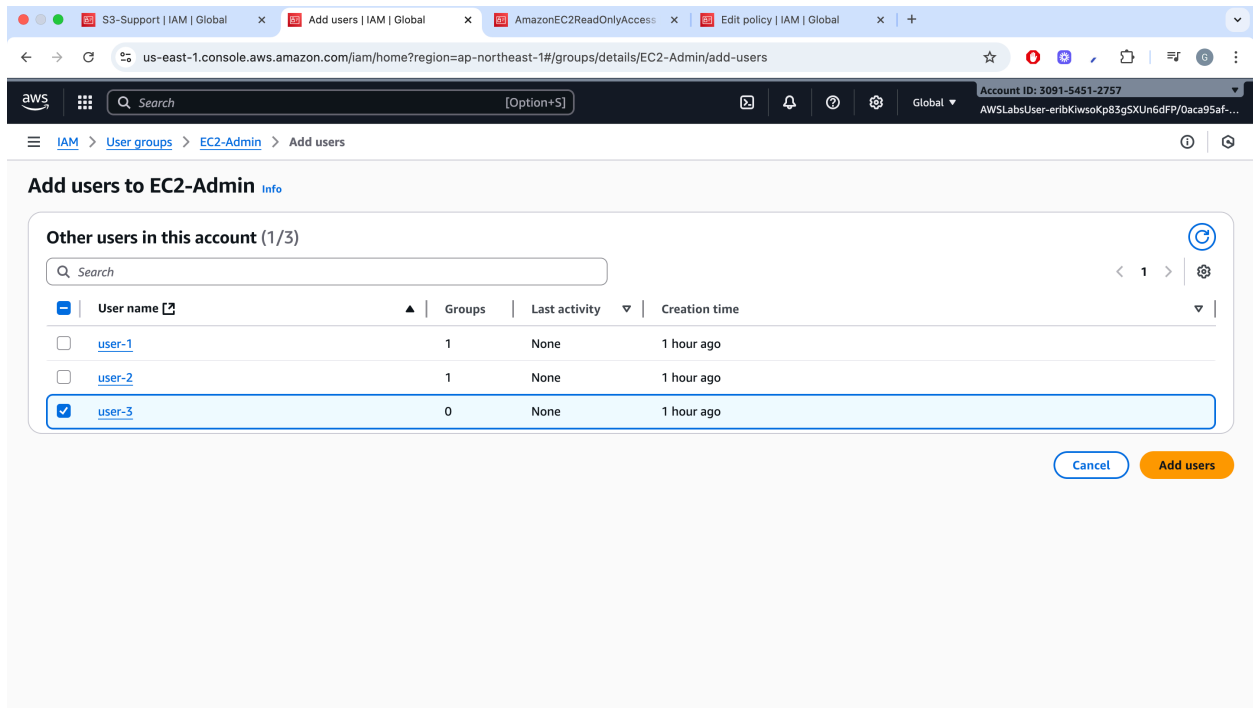


Figure 12: EC2-Admin group overview (source: personal collection)

Task 3: Sign-in and Test Users

- Step 1: Access the IAM users Sign-in URL
- Step 2: Open a private window in your browser
- Step 3: Sign-in as user-1, access the s3bucket, and the EC2 instance (this should be restricted).
- Step 4: Sign-in as user-2, access the EC2 instance, and attempt to stop it.
- Step 5: Sign-in as user-3, access the EC3 instance again, and stop the instance.

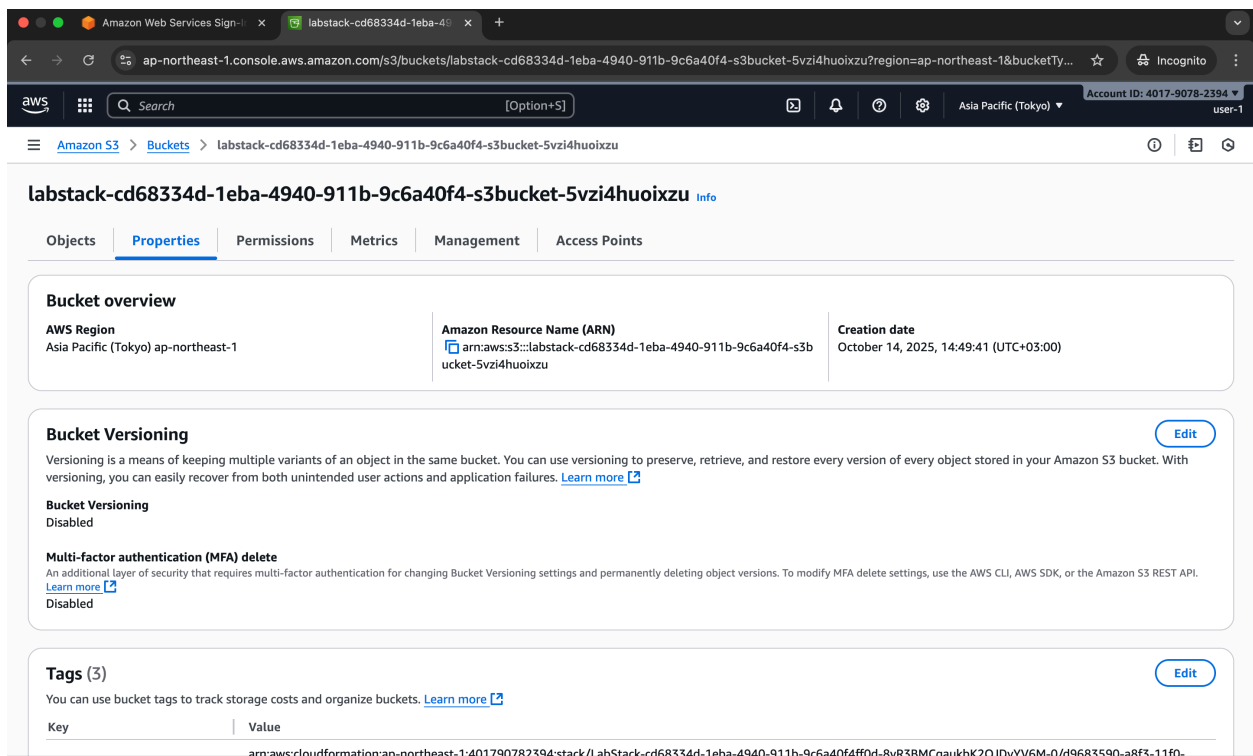


Figure 13: user-1 accessing the S3 bucket (source: personal collection)

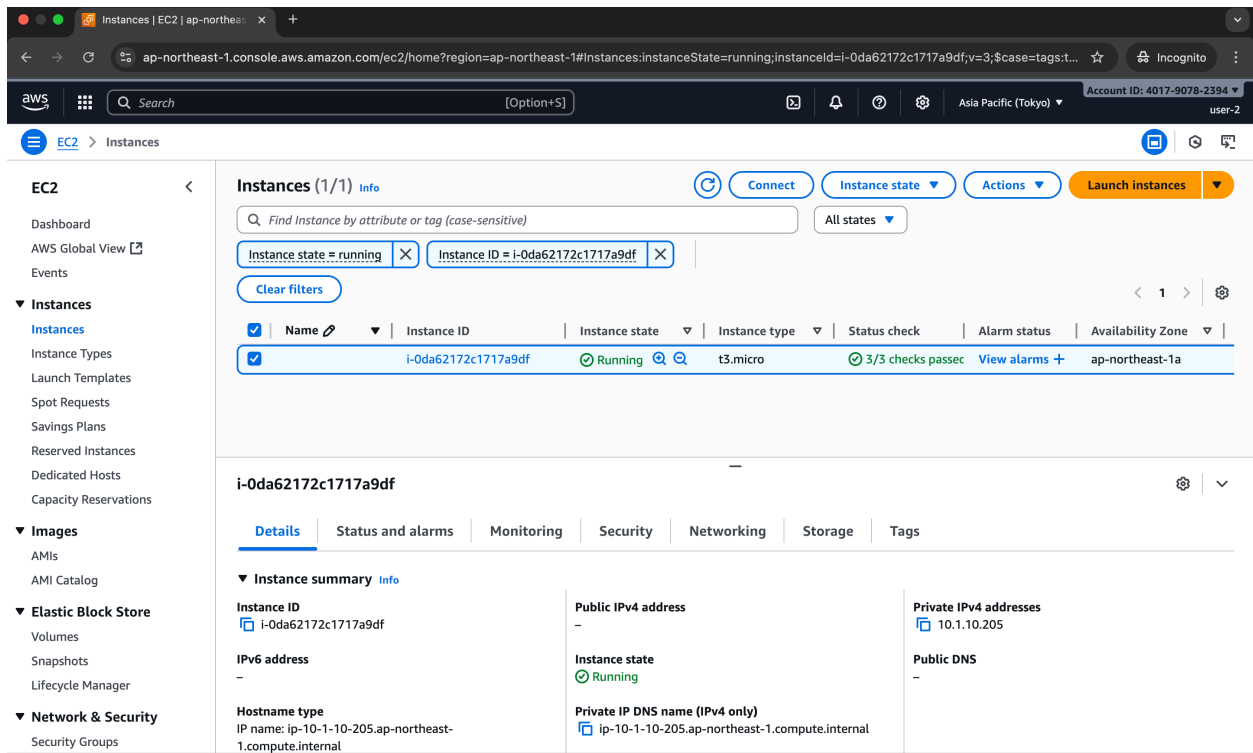


Figure 14: user-2 accessing the EC2 instance (source: personal collection)

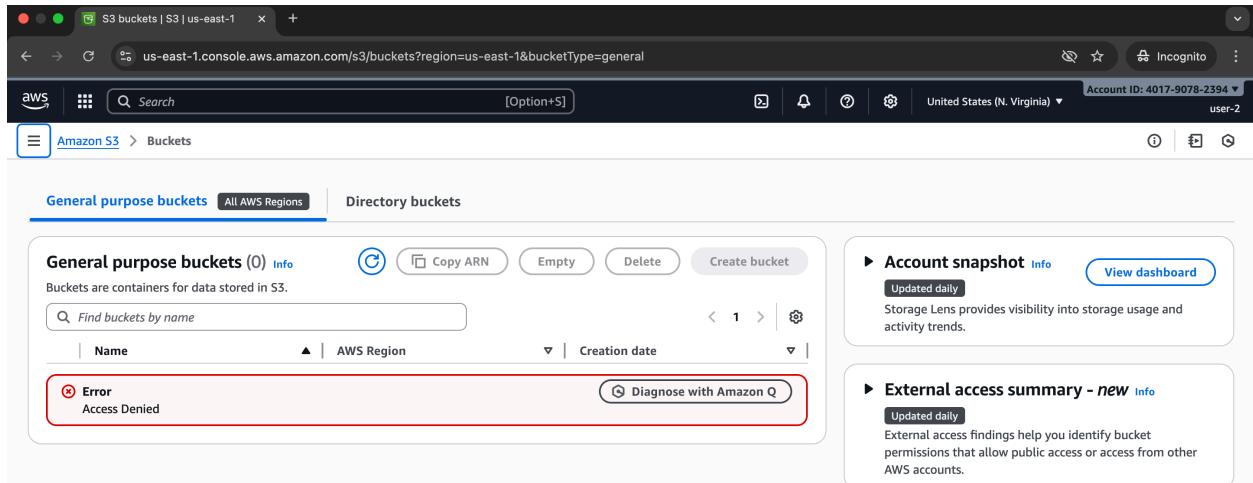


Figure 15: user-2 unable to access the S3 bucket (source: personal collection)

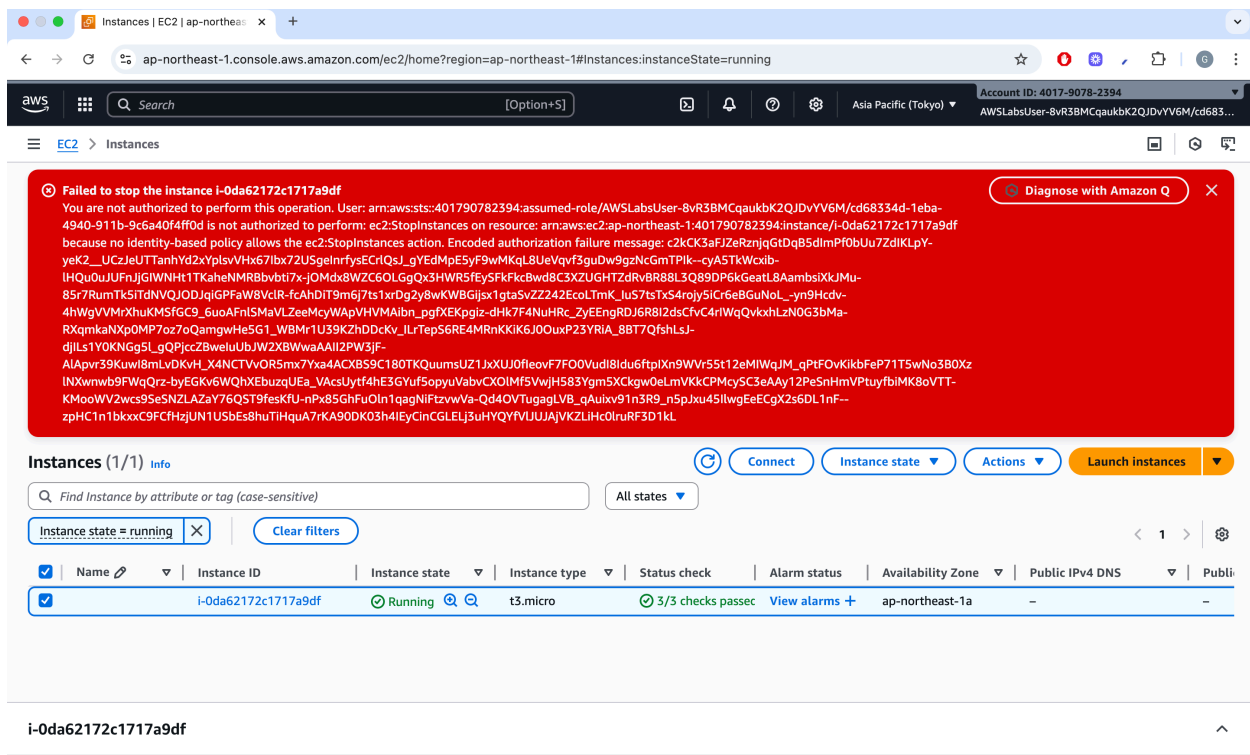


Figure 16: user-2 unable to stop the EC2 instance (source: personal collection)

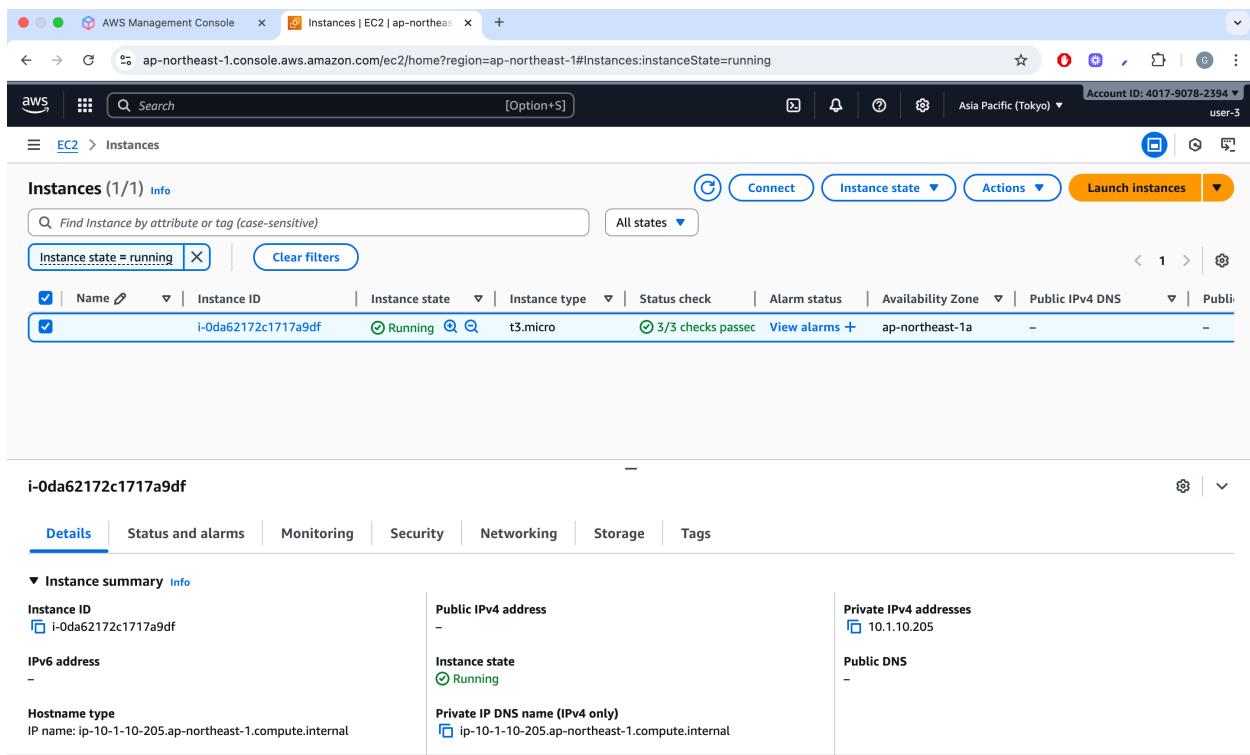


Figure 17: user-3 accessing the EC2 instance (source: personal collection)

Successfully initiated stopping with skip OS shutdown of i-Oda62172c1717a9df

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 address
i-Oda62172c1717a9df	i-Oda62172c1717a9df	Stopping	t3.micro	3/3 checks passed	View alarms	ap-northeast-1a	-	-

i-Oda62172c1717a9df

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info

Instance ID i-Oda62172c1717a9df	Public IPv4 address -	Private IPv4 addresses 10.1.10.205
IPv6 address -	Instance state Stopping	Public DNS -
Hostname type IP name: ip-10-1-10-205.ap-northeast-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-1-10-205.ap-northeast-1.compute.internal	

Figure 18: user-3 able to stop the EC2 instance (source: personal collection)

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

Stopped

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 address
i-Oda62172c1717a9df	i-Oda62172c1717a9df	Stopped	t3.micro	-	View alarms	ap-northeast-1a	-	-

i-Oda62172c1717a9df

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info

Instance ID i-Oda62172c1717a9df	Public IPv4 address -	Private IPv4 addresses 10.1.10.205
IPv6 address -	Instance state Stopped	Public DNS -
Hostname type IP name: ip-10-1-10-205.ap-northeast-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-1-10-205.ap-northeast-1.compute.internal	

Figure 19: EC2 instance stopped (source: personal collection)

Conclusion

In this lab, I explored how to add users to existing user groups in AWS. I managed access and permissions for AWS services using IAM. I then logged in with IAM-created users and observed how IAM policies affected access to specific services.

Reference

Amazon Web Services. (n.d). *Introduction to AWS Identity and Access Management (IAM)*.

Retrieved from <https://lab.builder->

[labs.skillbuilder.aws/sa/lab/arn%3Aaws%3Alearningcontent%3Aus-east-1%3A470679935125%3Ablueprintversion%2Fspl-66%3A3.1.21-57d5c391/en-US](https://lab.builder-labs.skillbuilder.aws/sa/lab/arn%3Aaws%3Alearningcontent%3Aus-east-1%3A470679935125%3Ablueprintversion%2Fspl-66%3A3.1.21-57d5c391/en-US)